Muhammad Usman Hanif

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Education

Comsats University Islamabad, Lahore Campus *Lahore*, Pakistan
Master of Science in Computer Science Sep 2022 – Jul 2025
Thesis: Anomaly Prediction and Analysis Using Deep Learning-Based Techniques
Subjects Studied: Data Mining, Big Data, Pattern Recognition, Artificial Neural Networks

University of Sargodha *Sargodha*, Pakistan
Bachelor of Science in Computer Science Nov 2016 – Oct 2021
Final Year Project: Web-Based House Rental System
Relevant Subjects: Data Structure, Algorithm Analysis, Database Systems, Artificial Intelligence

Experience

Comsats University Islamabad, Lahore Campus *Lahore,* Pakistan
Visiting Lecturer Sep 2025 – Jan 2026

- Taught Programming Fundamentals and Object-Oriented Programming to undergraduate students
- Helped students apply algorithms and programming concepts based on real-world scenarios.
- Improved presentation, communication, critical thinking, problem-solving, and programming skills.

Comsats University Islamabad, Lahore Campus *Lahore,* Pakistan
Master's Degree Research Work Jan 2024 – Jul 2025

- Implemented a deep learning technique to analyze CCTV videos, achieving 81% accuracy in predicting potential anomalous events before they occur.
- Performed data preprocessing pipeline, model evaluation, and improved predictive accuracy of 26% through hyperparameter tuning and architecture optimization.

Certifications

- **Communication and Soft Skills** – *Digiskills.pk* (*link*)
- **Freelancing** – *DigiskilsI.pk* (*link*)
- **E-Commerce Application Development** – *Punjab Board of Technical Education, Lahore* (*link*)

Professional Skills & Interests

- **Programming:** Python (NumPy, Pandas, Scikit-learn), OOP, JavaScript, C, C++
- **Machine Learning & Deep Learning:** Scikit-learn, TensorFlow, Keras, Model Training, Optimization, Fine-tuning
- **AI Engineering:** Large Language Models (LLMs), RAG, Prompt Engineering, Embeddings, Basic Agentic Workflows
- **Deployment & Integration:** FastAPI, REST APIs, CI/CD Basics, Model Serving Basics, Deployment Fundamentals
- **Databases:** SQL, MySQL, PostgreSQL, NoSQL, Database Design, Data Modeling, Query Optimization